Jie Zhang

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Education

Shanghai Jiaotong University Shanghai, China
Master of Engineering, Control Engineering *September 2016 - June 2019*
Concentration: Machine Learning and Computer Vision

Southeast University Nanjing, China
Bachelor of Engineering, Control Engineering *September 2012 - June 2016*

Skills

Programming/Scripting Languages: C++, Python, Perl, Bash, C, C#
Technologies/Environment: Git, CMake, Windows, Linux, LLVM, OpenGL, Tensorflow, Pytorch, Pandas
Languages: English (IETS 7.5), Mandarin

Experience

Intel Corporation Shanghai, China
Software Engineer - Intel Architecture Graphics and Software (IAGS) *June 2019 - present*

- Enhance the SYCL backend for llama.cpp and implement parts of SYCL backend adaptors, and support development process for the SYCL backend for deep learning frameworks (Tensorflow, Pytorch).
- Support test activities for the Data Parallel C++ compiler within 12 Intel oneAPI package releases. Provide detailed analysis on failures and triage over 900 issues to correct compiler components or dependency libraries
- Refine and optimize testing code based on DPC++ ABI and specification changes. Provide fixing patches and suggestions to 3rd party applications and increase application compatibility with the latest oneAPI product
- Take full charge of designing and developing DPC++ testing suites, including thread safety, dynamic linking support, and SYCL conformance test suite improvement. Implement a readable and extensible testing framework to support further compiler development
- Ensure DPC++ quality on the latest integrated and discrete graphic cards, including Sierra Forest, Lunar Lake, Arrow Lake, Alchemist, Battlemage, and other unreleased platforms. Set up and maintain the testing environment for new platforms, triage unique issues found on these platforms, and aid in fixing DPC++ compiler issues

Intel Corporation Shanghai, China
Software Engineer Intern - Intel Architecture Graphics and Software (IAGS) *March 2019 - June 2019*

- Developed code coverage tool for Data Parallel C++ compiler: (a) Took full charge of developing the basic block coverage function for LCOV. (b) Generated code coverage reports with function, line, and basic block coverage based on the current testing plan, and validated the correctness

Projects

Automatic Test Generator for Data Parallel C++

- Improved current fuzz generator to generate legal DPC++ code, including vector type initialization and conversion, kernel code generation and submission, and result restoration and computation with thread ids
- Implemented automatic failure analysis by cross-validating binary outputs with different devices and different optimization levels & compilation methods, which can automatically reduce source code with code-reducing tools and generate small reproducers without manual efforts
- Automatically create trackers and dispatch to the correct component owners, and add small reproducers into the internal test base.

SYCL Conformance Test Suite Improvement

- Contributor of the official SYCL Conformance Test Suite and gatekeeper of the Conformance Test Suite
- Reviewed testing gaps between Khronos SYCL Conformance Test Suite (SYCL CTS) and SYCL Specification, and drafted the improvement plan on result verification, type coverage, functor support, and SYCL API coverage
- Improved test coverage of SYCL CTS. Designed and implemented utility functions to refine result verification. Cover testing gaps in Khronos SYCL Conformance Test Suite (SYCL CTS) and SYCL Specification
- Support SYCL conformance certification. Improve compilation and execution efficiency by refactoring heavy cases

Publications

Zhang Jie, Xu Yi, et al. Geometric Constrained Joint Lane Segmentation and Lane Boundary Detection. 15th European Conference on Computer Vision (A *top-tier computer vision and machine learning conference*), 2018